Question:
A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K_4 Fe (CN)_6. After filtering, the precipitate is dried and weighed. The reactions which occur are ZnS + 2HCl \rightarrow ZnCl_2 + H_2 S 2ZnCl_2 + K_4 Fe (CN)_6 \rightarrow Zn_2 Fe (CN)_6 + 4 KCl If a 2 gram sample of rock yields 0.969 gram of Zn_2Fe(CN)_6, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn_2 Fe (CN)_6 = 342.6.
Answer Choices:
A. 10 % Zn
B. 22 % Zn
C. 20 % Zn
D. 19 % Zn
E. 15 % Zn
F. 18 % Zn
G. 16 % Zn
H. 14 % Zn
I. 12 % Zn
J. 25 % Zn

Reasoning + Answer:
Let's solve the problem step by step:

We are given:

- A 2 gram sample of rock yields 0.969 gram of ZnâĤĤFe(CN)âĤĨ.
- We are to find the **percentage of Zn** in the original rock sample.
- We are given:
- Atomic weight of Zn = 65.4
- Molecular weight of ZnâĤĤFe(CN)âĤĨ = 342.6

We are told that ZnS is dissolved by HCl and then ZnÂ²âģº is precipitated as ZnâĤĤFe(CN)âĤĨ.

From the chemical equations:

1. ZnS + 2HCl âĨĴ ZnClâĤĤ + HâĤĤS
2. 2ZnClâĤĤ + KâĤĦFe(CN)âĤĨ âĨĴ ZnâĤĤFe(CN)âĤĨ + 4KCl

So, 2 moles of ZnClâĤĤ produce 1 mole of ZnâĤĤFe(CN)âĤĨ.

Thus, 1 mole of ZnâĤĤFe(CN)âĤĨ contains **2 moles of Zn**.

So, we will calculate the mass of Zn in 0.969 g of ZnâĤĤFe(CN)âĤĨ, then use that to find the % Zn in the original rock sample.

Step 1: Moles of ZnâĤĤFe(CN)âĤĨ formed

Molar mass of ZnâĤĤFe(CN)âĤĨ = 342.6 g/mol

Moles of ZnâĤĤFe(CN)âĤĨ = mass / molar mass = 0.969 / 342.6

= 0.002829 moles

Step 2: Moles of Zn in the precipitate

Each mole of ZnâĤĤFe(CN)âĤĨ contains 2 moles of Zn

So, moles of Zn = 2 ÃĹ 0.002829 = 0.005658 mol

Step 3: Mass of Zn

Mass of Zn = moles ÃĹ atomic mass = 0.005658 ÃĹ 65.4

= 0.3704 g (approximately)

Step 4: Percentage of Zn in the rock sample

Total sample mass = 2 g

% Zn = (mass of Zn / total sample mass) ÃĹ 100
= (0.3704 / 2) ÃĹ 100
= 18.52%

So, the percentage of Zn in the sample is approximately **18.5%**

Looking at the choices:

A. 10% Zn
B. 22% Zn
C. 20% Zn
D. 19% Zn
E. 15% Zn
F. 18% Zn
G. 16% Zn
H. 14% Zn
I. 12% Zn
J. 25% Zn

Closest value to 18.5% is **19%** (D) âĢĶ though 18% is also close.

But letâĢĻs verify if we did it accurately.

Recalculate step